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NITI Aayog

DRAFT

SECTOR AT A GLANCE

Forestry

Agriculture and Allied Sectors in India

Sector at a Glance

Forestry

Agriculture and Allied Sectors in India

The **'Sector-at-a-Glance'** provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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In India, forests are treated as social, environmental, and commercial resources with more than 300 million people depending fully or partially on forests for their livelihoods. Forestry and logging products such as industrial wood, fuelwood, and Non-Timber Forest Products (NTFPs) contribute to approximately 8.7% of the total Gross Value Added (GVA) from the agriculture and allied sectors. NTFPs alone have the potential to contribute to about 20% to 40% of the annual income of forest dwellers, a predominantly tribal population.

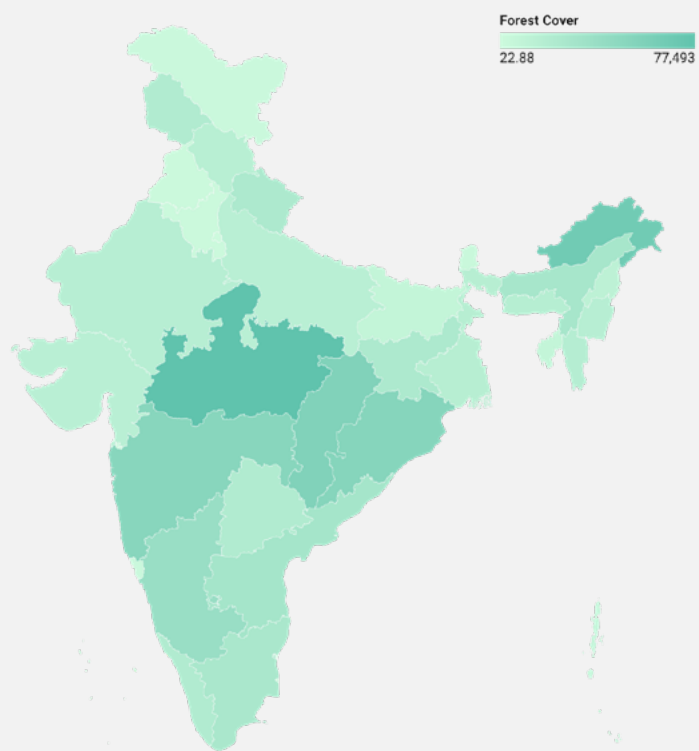
Promoting NTFP cultivation in agroforestry systems, sustainable management and trade practices, and increased investments from the government in the form of national and state-level policy reforms could serve as significant tailwinds for this sector. Although there has been significant development in NTFP-related policies, challenges pertaining to sustainable production and management still persist. However, the National Afforestation Programme, Intensification of Forest Management Scheme, and the Sub-Mission on Agroforestry and National Agroforestry demonstrate the government's commitment to addressing these challenges and promoting the sector's growth.

India's Forest Cover

India's forested area spans 7,13,789 square kilometres, covering **21.7% of the nation's entire geographical expanse**. From 2019 to 2021, India saw an increase of 1,540 square kilometres in its forest cover,¹ making forestry the second-most prominent land utilisation sector after agriculture.

Madhya Pradesh (77,493 sq. km) holds the foremost position in terms of forest coverage followed by Arunachal Pradesh (66,431 sq. km), Chhattisgarh (55,717 sq. km), Odisha (52,156 sq. km) and Maharashtra (50,798 sq. km). In terms of forest cover as a percentage of total geographical area, the top five States are Mizoram (84.5%), Arunachal Pradesh (79.3%), Meghalaya (76.0%), Manipur (74.3%) and Nagaland (73.9%).

Figure 1: Total Forest Cover by State, 2021 (Square Kilometres)



Source: India State of Forest Report, 2021

1 Ministry of Environment, Forests, and Climate Change. (2021). *India State of Forest Report*.

Economic Activity Supported by Forests

The forestry and logging sector contributes **approximately 7.9% to India's Gross Value Added (GVA)** from agriculture and allied sectors (2021-22, Current Prices).² Roughly **26% of the country's 6,50,000 villages are said to be forest-fringe villages**, which rely somewhat on forests for their sustenance and livelihood. These ~170,000 villages are **home to an estimated 22% of the Indian population** (29.4 crore people).³



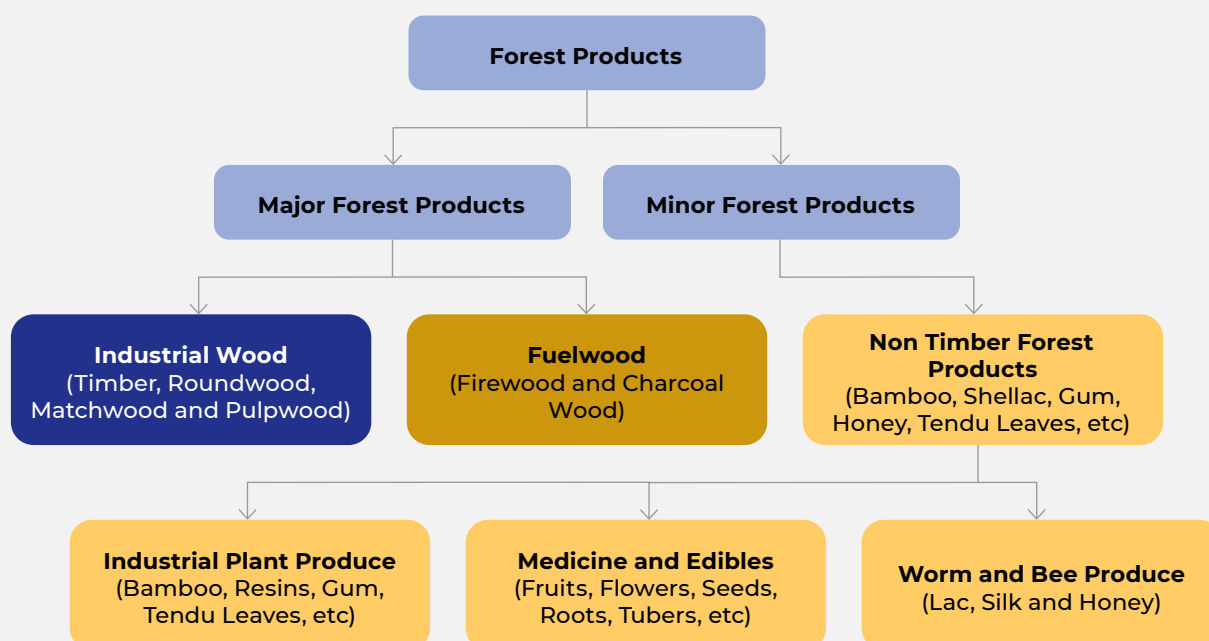
India's National Accounts Statistics document the following **forest-based economic activities**:

1. **Forestry** (e.g. planting and conservation of forests, gathering of forest products, charcoal burning carried out in forests etc.)
2. **Logging and transportation** (e.g. felling and rough-cutting of trees, hewing or rough-shaping of poles and blocks) and transportation of forest products to the sale depots/assembly centres etc.),
3. **Farmyard wood** (industrial wood and fuelwood collected by primary producers from trees outside regular forests)⁴

Forest-based activities result in products that can be classified into two broad groups:

1. **Major products** comprising of industrial wood and fuelwood
2. **Minor products** comprising wild growing forest material such as bamboo, lac, honey, resin, gum, tendu leaves etc. (most NTFPs)

Figure 2: Classification of Indian Forest Products



Source: Department of Fisheries Annual Report 2022-23

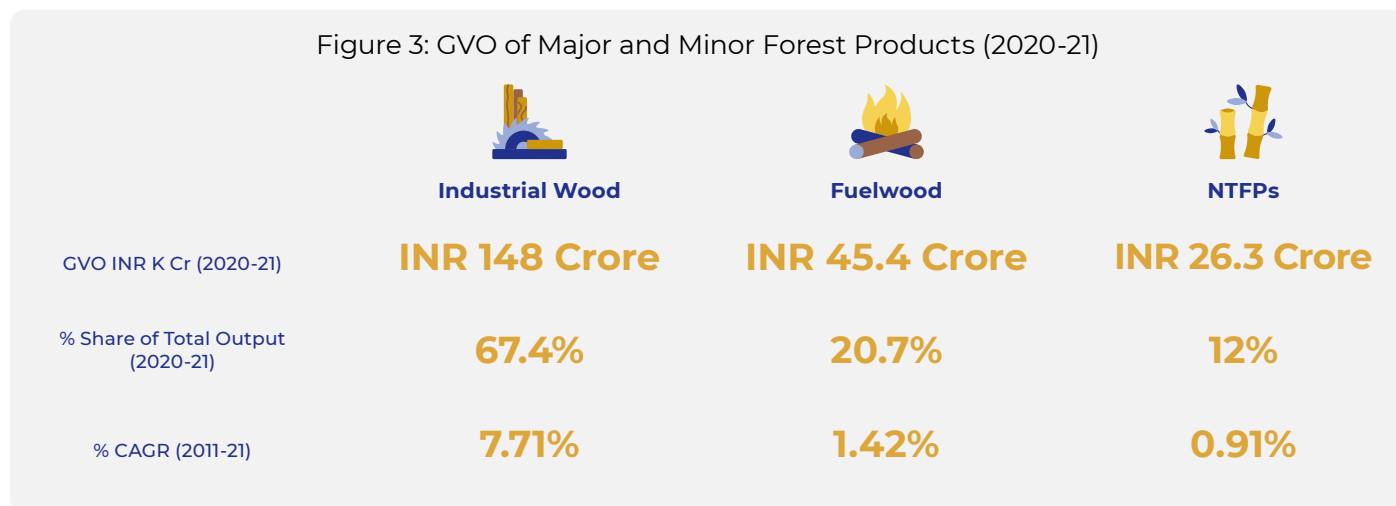
² Ministry of Agriculture & Farmers Welfare. (2022). *Agriculture Statistics at a Glance 2022*.

³ Ministry of Environment, Forests, and Climate Change. (2019). *India State of Forests Report*.

⁴ Ministry of Statistics and Programme Implementation. (2007). *National Accounts Statistics — Sources and Methods, 2007*.

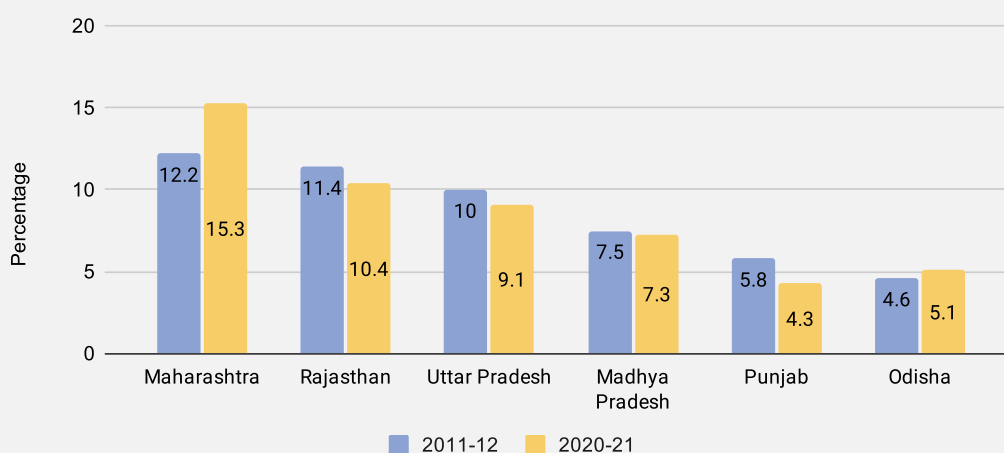
The data below shows the Gross Value of Output (GVO) for major and minor forest products as of 2020-21.⁵

Figure 3: GVO of Major and Minor Forest Products (2020-21)



In 2020-21, Maharashtra and Rajasthan accounted for over 25% of this total output (15% and 10% respectively), followed by Uttar Pradesh (9%), Madhya Pradesh (7%), Odisha (5%), and Punjab (4%).⁵

Figure 4: % Share in Forest Product Output for Top Five States (2011-12 and 2020-21)



Sources: Ministry of Statistics and Programme Implementation

Agroforestry and Non-Timber Forest Products

Agroforestry, a land use system combining tree-growing with agricultural practices, currently **fulfils approximately 72% of India's total fuelwood demand**. This is significant as approximately **47% of the country's rural population** meets their household cooking **requirements from fuelwood** obtained from forests.⁶ Agroforestry also addresses the rural population's other needs such as food, fruits, fodder, fibre, and medicine, etc.⁷ Domesticating NTFP trees grown as part of agroforestry systems has shown to offer substantial employment opportunities for rural youth for majority of the year in comparison to agriculture alone.⁸ NTFPs have the potential to fulfil **55% of the total employment in the primary forest sector**, and their impact is multiplied through downstream processing and trading of specific products.

5 Ministry of Statistics and Programme Implementation. (2023). *State-wise and Item-wise Value of Output from Agriculture, Forestry and Fishing, Year: 2011-12 to 2020-21*.

6 Ministry of Statistics and Programme Implementation. (2023). *Multiple Indicator Survey in India (2020-21)*.

7 Sahoo et al. (2019). *Multifunctional agroforestry systems in India for livelihoods*.

8 Subbulakshmi et al. (2018). *NTFP based Agroforestry to Sustain Income and Employment Generation Activities of Arid Regions of Rajasthan*.

Sub-sectoral Trends

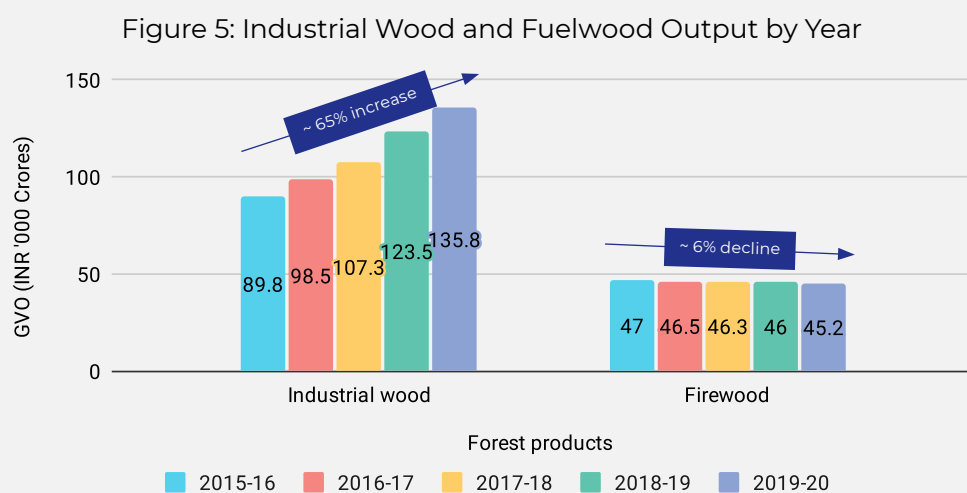
This section of the report discusses trends in the production and exports of forest-based products, highlighting some key data points and facts pertaining to:

- Industrial wood and fuelwood
- NTFPs such as bamboo, guar gum, shellac, honey, and medicinal plants

Industrial Wood and Fuelwood⁵

Industrial Wood

Wood-based industries such as saw-milling, paneling, carpentry, packaging, paper manufacturing, etc. are growing at a fast rate, leading to a widening gap between the demand and supply of raw wood material. To close this gap, India's agroforestry ecosystem has been redesigned, where popular wood tree species have been introduced into farm lands traditionally used solely for agricultural crops.⁹



The output of the forestry and logging sub-sectors has shown an increase from INR 74,300 crores in 2011-12 to INR 1,48,000 crores in 2020-21 (constant 2011-12 prices). In particular, the output of industrial wood increased from INR 89,000 crores in 2015-16 to INR 1,48,000 crores in 2021 — a ~65% growth. The **percentage share of Industrial wood in the total output of forest products increased** from 49.9% in 2011-12 to 67.4% in 2020-21.

Fuelwood

Fuelwood remains a readily available and affordable source of energy for many households in rural India, including for cooking and heating. Over the past three decades, the demand for fuelwood has been primarily met by forests, farmlands, and community lands. The total fuelwood extraction from forests was 85.3 million tonnes as of 2019 with **agroforestry on farmlands meeting 40% of the country's energy needs**.¹⁰

⁹ Kumar Meena et al. (2023). *Growth of Wood Based Industry through Industrial Agroforestry in India*.

¹⁰ Nagar et al. (2021). *Fuelwood and fodder consumption patterns among agroforestry-practising smallholder farmers of the lower Himalayas, India*.

Fuelwood output has remained more or less constant between 2011-12 to 2020-21, ranging from INR 48,100 crores to INR 45,400 crores (5.6% decline). However, the **percentage share of fuelwood in the total output of forest products decreased** from 32.3% in 2011-12 to 20.7% in 2020-21.

Bamboo

Bamboo, a versatile and fast-growing grass, plays a crucial role in both India's agricultural sector and forest ecosystem, particularly when incorporated into agroforestry practices.. India is the second largest cultivator of bamboo with 136 varieties being cultivated across 15.7 million hectares.¹¹ Many resourceful bamboo species such as B. bambos are cultivated in the country's northeastern regions.¹²

402 MT*

Estimated green bamboo culms (2021)

USD 544K

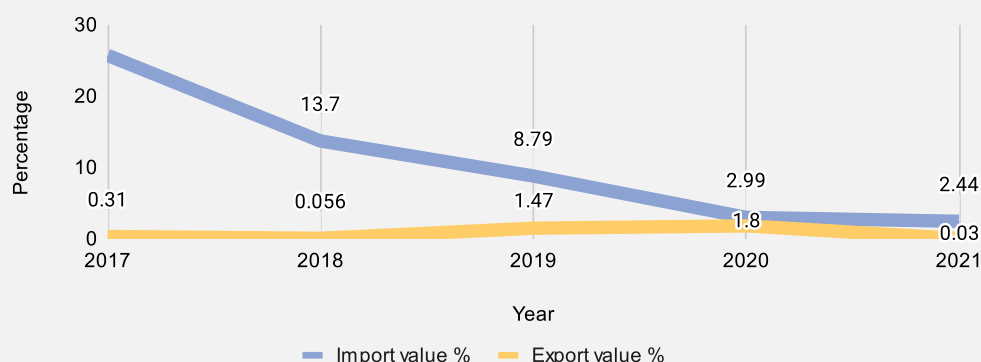
Total import value in 2021

Estimated green bamboo culms (2021)¹³

Note*: MT refers to metric tonnes

While the paper industry remains a major consumer of bamboo, its applications extend far beyond to the handicrafts and other bamboo-based industries. Being a sustainable alternative to timber, it is frequently used in construction as well. Since the domestic cultivation of bamboo has grown, **India's share of global bamboo pulp imports has decreased** over the past five years from 25.7% in 2017 to 2.4% in 2021. However, India does not export much bamboo pulp yet.¹⁴

Figure 6: India's Share of Global Bamboo Pulp Exports and Imports by Year



Source: OEC, Trade Data for Pulp of Bamboo

Guar Gum

Cyamopsis tetragonoloba, an annual plant that is a source of guar gum, is grown as a food crop for animals, and is one of the most important trees for agroforestry in India's dry regions.

India is the **world's leading exporter of guar gum**, a water-soluble polysaccharide extracted from the guar bean. The country exported 4,06,513 MT of guar gum worth INR 4,944 crores (USD 617 million) in 2022-23. This crop is primarily cultivated in Rajasthan, Gujarat, Haryana, Punjab, Uttar Pradesh, Madhya Pradesh, Tamil Nadu, Maharashtra, Karnataka, and Andhra Pradesh.¹⁵

¹¹ National Bamboo Mission and Department of Agriculture, Cooperation & Farmers Welfare. (2021). *National Consultation on Opportunities and Challenges for Bamboo in India*.

¹² Tewari et al. (2015). *Bamboo Based Agroforestry Systems*.

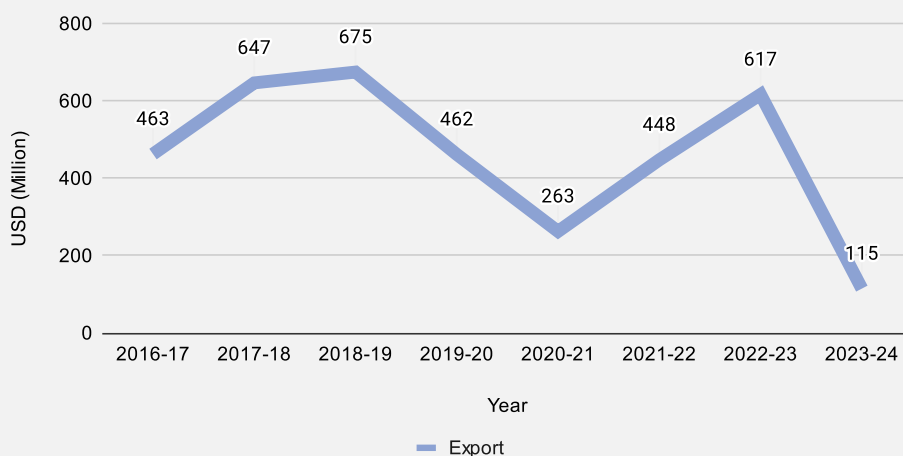
¹³ Ministry of Information and Broadcasting. (2021). *India State of Forest Report, 2021*.

¹⁴ The Observatory of Economic Complexity. *Pulp: of bamboo*. Retrieved 3 October, 2023.

¹⁵ Agriculture and Processed Food Products Export Development Authority. *Guar gum*. Retrieved 3 October, 2023.

The total guar seed production in India as of 2014-15 was 1,959 thousand MT. The production and export of guar seeds in India have increased significantly over the past decade due to the high demand for guar products from importing countries, especially the United States of America, whose oilfield sector uses them for extracting shale oil and gas.¹⁶

Figure 7: Guar Gum Export and Import by Year

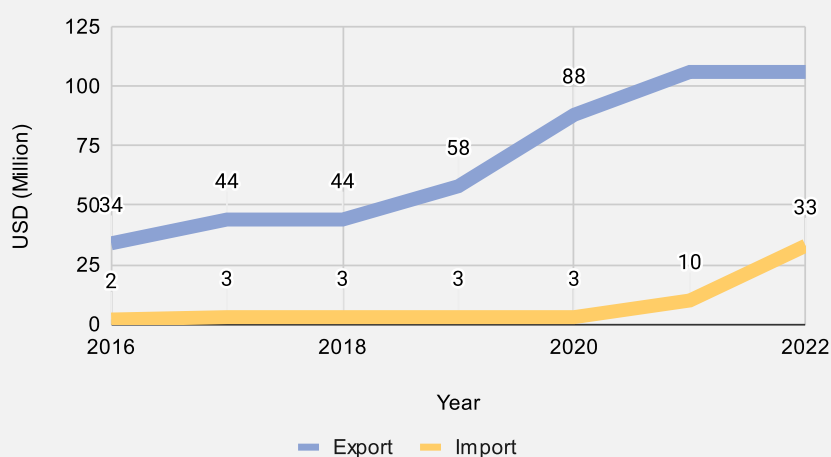


Source: Ministry of Commerce and Industry

Shellac

Shellac is a natural organic resin that has a wide range of uses from pharmaceuticals, to confectionaries, food coatings and electrical appliances. India has explored innovative agroecosystems that integrate lac cultivation in sugarcane neighbourhoods, providing natural host environments for the lac insect to increase overall lac production.¹⁷

Figure 8: Shellac Export and Import by Year



Source: Ministry of Commerce and Industry

India is a major producer of shellac- and lac-based products, the superior quality of which are internationally recognised. These products play an important role in the lives of millions of rural and forest-dependent people.¹⁸

¹⁶ National Bulk Handling CommWatch. (2015). *Seasonal Commodity Insight: Indian Guar Industry and Trade*.

¹⁷ Jaiswal et al. (2019). *Lac-Based Agroforestry System for Degraded Lands in India*.

¹⁸ Shellac and Forest Products Exports Promotion Council. *Annual Report 2017-2018*. Retrieved 3 October, 2023.

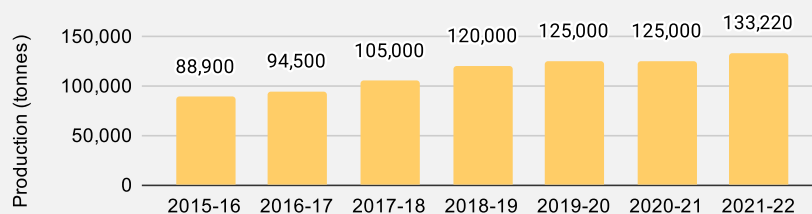
Shellac and lac-based product exports amounted to **USD 106 million** during **2021-22, a 20.5% increase** from USD 88 million from the previous year. India has recorded a consistent growth in its exports of shellac and lac-based products over the past seven years. **The average percentage growth rate of shellac exports from India has been ~22%** over the 2016-22 period.¹⁹

Honey

Beekeeping stands out as a crucial and sustainable endeavor that seamlessly blends forestry and agricultural practices. Beyond the allure of high-value honey and other hive products, this practice plays a pivotal role in advancing forest conservation. Moreover, it serves as a valuable avenue for farmers to generate supplementary income, thereby underscoring its economic and environmental significance.²⁰

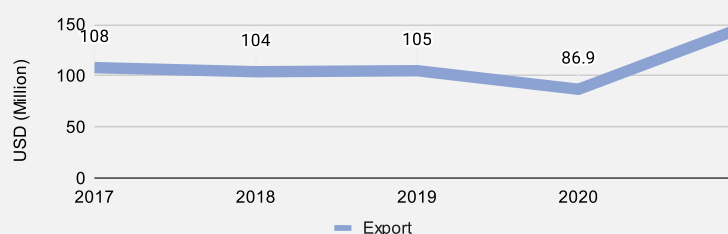
According to the National Bee Board (NBB), honey production has increased in India over the past few years, from 88,900 tonnes in 2015-16 to 115,000 tonnes in 2018-19, growing at **an average annual rate of 5%**.²¹

Figure 9: Honey Production in India by Year



Source: National Bee Board

Figure 10: Honey Export and Import by Year



Source: OEC Trade Data for Honey

In 2021, India exported USD 146 million in honey, making it the sixth largest honey exporter in the world. The country's honey exports have been steadily increasing from 2017 to 2022 at an average growth rate of approximately 12%. In comparison, the country's honey import has consistently remained low and currently accounts for 0.3% of the total value of honey imported globally. The United States of America is the biggest importer of Indian honey (78%), followed by Saudi Arabia and United Arab Emirates (4% each), and Canada and Nepal (2% each).²²

India's honey market was valued at INR 2,330 crores in 2022 and is expected to grow at a CAGR of 8.4% to INR 3,880 crores by 2028.²³

¹⁹ Ministry of Commerce and Industry. Department of Commerce: Monitoring Dashboard. Retrieved 3 October, 2023.

²⁰ Ministry of Micro, Small & Medium Enterprises. *Operational Guidelines For The Pilot Projects Of Beekeeping Activity Under Agro-Based & Food Processing Industries (ABFPI) Vertical Of Gramodyog Vikas Yojana (GVY)*. Retrieved 3 October, 2023.

²¹ Economic Advisory Council to the Prime Minister Government of India. (2019). *Report of the Beekeeping Development Committee*.

²² The Observatory of Economic Complexity. (n.d.). *Natural Honey*. Retrieved 15 October, 2023.

²³ International Market Analysis Research and Consulting Group. (2022). *Indian Honey Market Report by Flavor (Multiflora Honey, Eucalyptus Honey, Ajwain Honey, Sidr Honey, and Others), Seasonality (Autumn and Spring Season, Winter Season, Summer and Monsoon Season), Distribution Channel (Business to Consumer, Business to Business), and Region 2023-2028*.

Medicinal Plants

India is the second highest producer of medicinal plants in the world, after China. Medicinal plants are widely used by all sections of the population in India, and it has been estimated that, in total, over 7,500 species of plants are used by several ethnic communities.²⁴

The importance of medicinal plants has gained substantial attention in the country with the establishment of the AYUSH ministry. National AYUSH Mission has taken up various interventions to improve availability of medicinal plants and herbs in the country. In addition to this, CSIR-Central Institute of Medicinal and Aromatic Plants has been playing a key role in positioning India as a global leader in Medicinal and Aromatic Plants like mints, vetiver, and is playing an important role in commercialization of ayurveda based medicines.

According to the All India Trade Survey of Prioritised Medicinal Plants (2019), the demand for high-value medicinal plants increased by 50%, while their availability declined by 26%. The market for medicinal plants in India amounted to USD 56.6 million as of 2019 and is expected to grow at a CAGR of 38.5% by 2026.²⁴ With NTFPs accounting for 70% of India's forest product exports, the demand for phytochemicals is projected to increase creating new opportunities for trade.²⁵

Export Value of Herbs²⁶

USD 476.09 Mn

Export value (2017)

USD 630.09 Mn

Export value (2021)

India's share in the world medicinal plants exports has been growing along as the world market expands. India's medicinal plants registered a growth rate of 7.8% between 2017 and 2021. Psyllium, Turmeric, Zedovary Roots, and Sena Leaves account for approximately 85% of all Medicinal and Aromatic Plant exports. It is important to note that the demand for herbs/value-added extracts of medicinal herbs is gradually increasing in foreign countries, especially European nations and other developed countries. United States (34.9%), EU (18.66%) and UAE (5.52%) accounted for the highest quantity of herb imports from India in 2021.²⁷

Opportunities & Challenges

Opportunities

Opportunities within the forestry sector can be broadly divided into two types:

1. **Opportunities focused on growing agroforestry practices in India:** This includes efforts made by the government to incorporate more degraded land into agroforestry practices and identify products and species of plants that be grown as a part of the agricultural ecosystem.
2. **Opportunities focused on streamlining the management of NTFPs grown as a part of the agricultural ecosystem:** This includes initiatives that promote skilling and certification of farmers to improve post-harvest management of NTFPs and establishing fair trade practices to promote ease of sale.

²⁴ India Brand Equity Foundation. (2020). *High Demand for Medicinal Plants in India*. Retrieved 3 October, 2023.

²⁵ Shellac and Forest Products Export Promotion Council. *Panels: Herbs*. Retrieved 3 October, 2023.

²⁶ Pathak, N. & S. Agarwal. (2023). *AYUSH Exports: Regulatory Opportunities and Challenge in Key Markets. Research and Information System for Developing Countries*.

²⁷ Shellac and Forest Products Export Promotion Council. *Region-wise/panel-wise export data, 2009-18*.

Opportunities focused on growing agroforestry practices in India



Increasing area used for bamboo cultivation to meet growing the crop's growing domestic and global demand:¹²

As part of the Atmanirbhar Bharat Project, India aims to meet its bamboo paper and pulp demands domestically by improving bamboo production in the country. The National Bamboo Mission (NBM) plans to plant bamboo on an additional 1,05,000 hectares of land by ensuring sufficient stocks of high-quality planting materials. States with social, commercial, and economic advantages such as those in Northeastern India are the NBM's focus areas.²⁸



Promoting new types of agroforestry through the Sub-Mission on Agroforestry Scheme:²⁹

The Ministry of Agriculture and Farmers' Welfare has signed an agreement with the Central Silk Board (CSB) to promote agroforestry in the silk sector. The industry's and farmers' perspectives of agroforestry — as a sector that revolves around timber products — needs to evolve. While timber products have a long maturity, they generally cause delays in returns for farmers; thus, growing sectors like silk, lac, paper, pulp, etc., which ensure quick returns and meet industry requirements, are to be explored.³⁰



Promoting the cultivation and export of medicinal plants:

The Government of India has taken several measures to increase the production of medicinal plants. For instance, the National Medicinal Plants Board (NMPB) through a centrally sponsored scheme offers subsidy at the rate of 75%, 50%, and 30% of cost of cultivation is provided to 140 priority medicinal plants to farmers.³¹

Opportunities focused on streamlining the management of NTFPs



Providing training for sustainable harvesting of medicinal plants and other NTFPs:

Production of medicinal plants tend to be affected by overharvesting, pre-mature harvesting and damage to the source tree/plant. Therefore, it is important to train the harvesters in sustainable harvesting practices for medicinal plants & NTFPs. For example, Ecosystem Service Improvement Project being assisted by the World Bank and implemented by the Green India Mission of the State Forest Department of Madhya Pradesh has developed protocols for sustainable harvesting and is providing training to local villagers in this and other aspects.³²



Promoting industry-oriented skilling to improve the post-harvest management of bamboos:

Initiatives such as the Green Skill Development Programme aim to develop the skills of workers with a commitment to the sustainable development of bamboo through specialised courses on propagation, management, and marketing of bamboo and other NTFPs. Skill enhancement is an integral part of state-level initiatives such as the Assam Bamboo and Cane Policy, the Manipur Bamboo Policy, etc. Additionally, the NBM focuses on building traditional artisanship and imparting new skills to farmers.¹²

28 Sawarkar et al. (2020). Commercial clustering of sustainable bamboo species in India.

29 Press Information Bureau Government of India. (2021). Ministry of Agriculture and Farmers Welfare signs MoU with Central Silk Board, Ministry of Textiles for promotion of Agroforestry in Silk Sector.

30 Press Information Bureau Government of India. (2020). Webinar on Connecting Agroforestry Farmers to Industry, Retrieved 3 October, 2023.

31 National Medicinal Plant Board (2021). Ayush Next. Retrieved 20 October 2023.

32 Green India Mission, Madhya Pradesh Forest Department. (2021). Sustainable Harvesting Protocol for NTFPs and Medicinal Plants.



Prioritising the certification of NTFPs:

Such prioritisation will help minimise the ecological consequences of NTFP management and increase socio-economic benefits. The compliance of the certification scheme with the state, national, and international laws will help ensure the NTFPs are of the highest quality while serving as a quality assurance tool for consumers in the domestic and international market.³³



Ensuring better price realization for harvesters:

Can be achieved if Farmer Producer Organizations procure NTFPs collected from harvesters at market price while selling the collected NTFPs to large size-traders. This model has been found to be beneficial under the State Rural Livelihood Mission funded project for Sustainable Livelihoods Enhancement through NTFPs.³⁴

Challenges^{35,36,37,38}

Challenges with NTFPs grown as a part of agroecosystems can be broadly categorised into four groups:

1. **Production-level challenges** are associated with growing and harvesting NTFPs.
2. **Policy-level challenges** are associated with policy-level tailwinds and institutional frameworks that require further development for overall NTFP management.
3. **Value-chain-level challenges** are associated with limited technical skills, modern tools and facilities.
4. **Market-level challenges** are associated with a lack of market linkages that enable a fair price for farmers in domestic and international markets.

 Production level challenges	• Unsustainable MFP harvesting techniques leading to poor resource generation • NTFP raw materials gathered from government-owned lands making planning for sustainable production more difficult than on private lands due to unclear/complex access and collection rules • Small-scale production due to the rural population's lack of adequate skills and tools for collecting and processing NTFPs • Unpredictable productivity and over-exploitation of NTFP collectors since most NTFPs are seasonal
 Policy level challenges	• Difficulty in the development of policies and regulations due to the lack of a clear definition of NTFPs (inclusion/exclusion of products of plant/animal origin or wood products) • Requirement of an apex body for overall NTFP management and coordination between government agencies • Difficulty in assessing NTFPs' economic and environmental importance due to inadequate data

33 Pant et al. (2014). NTFP Certification in India: Prospects and Problems.

34 Atal Bihari Vajpayee Institute of Good Governance and Policy Analysis. (2020). Impact Assessment and Evaluation Study: Sustainable Livelihoods Enhancement Through NTFP Under MPSRLM.

35 Gramin Samassya Mukti Trust. (2021). *Non-Timber Forest Produce Collection, Processing and Market Cluster in Yavatmal District, Maharashtra*.

36 Indira Gandhi National Forest Academy. *Policy Issues in NTFP Management*. Retrieved 3 October, 2023.

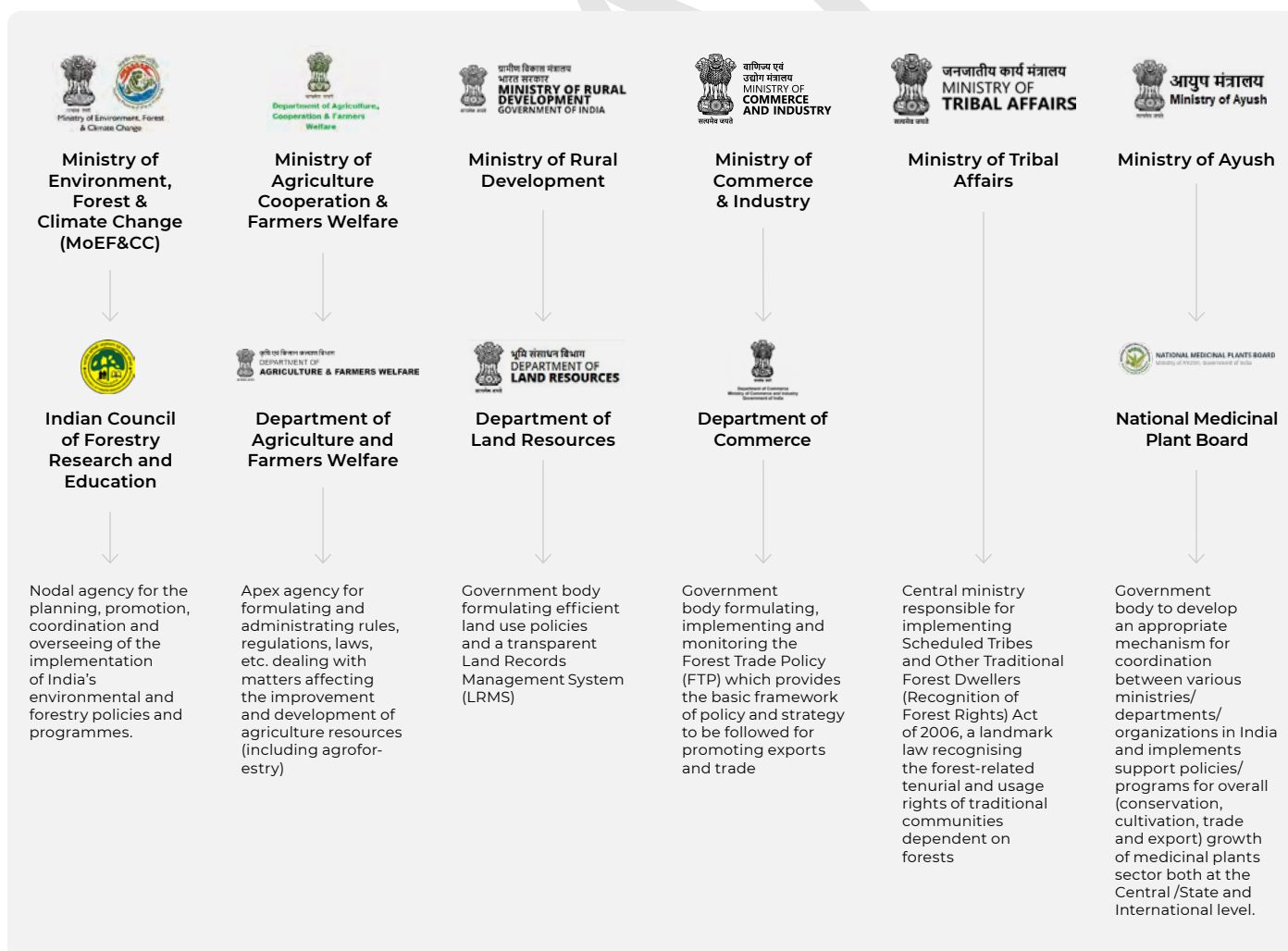
37 Bag et al. (2010). NTFP Policy Regime after FRA : A study in select states of India.

38 Food and Agriculture Organisation. *Marketing information systems for non-timber products*. Retrieved 3 October, 2023.

 Value chain level challenges	• Inequitable distribution of profits across the value chain due to the largely unorganized and informal operations of the sector • Irregularities in quality standards due to the absence of sustainable, scientific and non-destructive methods to collect and manage NTFPs • Lack of facilities for storage, processing, and distribution • Lack of tools, skills and knowledge in accessing government schemes and post-harvest management of NTFPs
 Market level challenges	• Poor awareness and lack of capacity of the local communities about the ownership, management and sale of NTFPs • Lack of financial support to primary collectors and their institutions in the form of revolving funds, open action systems and funds provided by the village forest councils • Lack of market linkages with NTFP collectors and local procurement centres at the haat-bazaar level • Lack of visibility regarding corporate buyers and e-commerce options for selling products

Government Interventions

Ministries and Departments



Policies, Schemes, and Initiatives

National Forest Policy (1988) Launched by the Government of India in 1988, the policy aims to ensure the sustainable management of forests and environmental conservation	Nagar Van Yojana Launched in 2015 by the Ministry of Environment, Forest and Climate Change, it aims to create urban forests in cities and towns	National Afforestation Programme Launched in 1976 by the Ministry of Environment, Forest and Climate Change, the programme aims to increase the forest cover in the country to 33% of its geographical area, improve the quality of existing forest and create employment opportunities
Green India Mission One of the eight missions of the National Action Plan on Climate Change, 2014, it aims to protect, restore and enhance India's diminishing forest cover	Intensification of Forest Management Scheme The scheme aims to improve the quality of existing forests by planting more trees, improving forest management and creating awareness about the importance of forests	Compensatory Afforestation Fund Management and Planning Authority Established in 2006 under the Forest Conservation Act, 1980, the fund compensates the loss of forest cover due to development projects
Sub-Mission on Agroforestry (SMAF) Launched in 2015, the National Mission for Sustainable Agriculture aims to promote the adoption of agroforestry by farmers, create awareness, and support research and development	National Agroforestry Policy Launched in 2014, the policy aims to promote agroforestry in India by integrating it with other land-use systems such as agriculture and forestry	National Bamboo Mission (NBM) Launched in 2006-07, the NBM was merged with the Mission for Integrated Development of Horticulture (MIDH) in 2022-23. The restructured NBM focuses on the complete bamboo value chain, linking growers to consumers, encompassing planting, facilities, skilled manpower, marketing, and brand building. Implemented in 24 States/UTs, it supports bamboo plantations and bio-energy units for government and private entities.
National Beekeeping and Honey Mission Launched in 2020, NBHM aims to enhance scientific beekeeping, boost honey production, and improve beehive product quality. It comprises three Mini Missions: (a) enhancing crop productivity through scientific beekeeping, (b) focusing on post-harvest management and infrastructure development for honey and hive products, and (c) promoting research and technology to adapt beekeeping for diverse regions and socio-economic conditions.		



Way Forward

 National- and Regional-level Actions to Promote Agroforestry	• To facilitate pricing, processing, procurement, credit, insurance, and marketing, and provide financial incentives to farmers, an agroforestry board — similar to the coconut and coffee boards — can be established under the National Agroforestry Policy. • To accelerate agroforestry efforts at a district level, a subject matter expert can be appointed by each Krishi Vigyan Kendra. • As recommended the by National Agroforestry Policy, a regional consortium/ network can be set up in joint partnership with the International Centre for Research in Agroforestry (ICRAF) and Asia Pacific Association of Agriculture Research Institutions (APAARI) to ensure policy advocacy, research collaboration, knowledge sharing and capacity building, • As suggested in the National Mission for Green India, independent research studies can be sanctioned and set up by government bodies to assess and scale up innovations in agroforestry.³⁹
 Enhancing Agroforestry to Support Local Livelihoods and Food Security	• As mentioned in the National Forestry Policy 1988, geographical and social context-specific silviculture and farming systems need to be designed to optimise food production and biodiversity conservation. • To encourage more farmers to take up NTFP plantation, recommendations have been made by the National Agroforestry Programme to raise capital through venture funds, loans, microfinancing and strong market linkages.⁴⁰ • Enable Farmer Producer Organizations to NTFP management as it gives bargaining edge to harvesters, leads to use of sustainable collection practices and allows harvesters to tap into capacity building schemes.
 Integration of Technology to Enhance Agroforestry Practices	• As per the Madhya Pradesh State Forest Policy, biotechnology techniques can be used to strengthen commercially important forest species and their specific qualities. • To diagnose land management problems and design subsequent agroforestry solutions, an integration of traditional systems and modern technology is recommended by the state forest policies of Assam, Madhya Pradesh, Chhattisgarh and others. • To monitor the implementation of forestry plans and schemes, a Geographical Information System (GIS) is being set up by the Government of Chattisgarh to promote better governance and dealing of forest products.

Recommended Readings

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